# Week 01: Sets and Counting Principles

**Exercise 1.** Let the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and its subsets:

- $A = \{1, 2, 4, 8\}$
- $B = \{x \in U \mid x \text{ is odd}\}$
- $C = \{5, 6, 7, 8\}$

Find the following sets or calculate the following values:

(a) $A \cup B$

(b) $A \cap B$

(c) $A \setminus B$

(d) $A^c$

(e) $(A \cup B) \setminus C$

(f) $A^c \cup C$

(g) $A^c \cap C$

(h) $|A|$, $|A| - |B|$, $|A \setminus C|$

**Exercise 2.** A person wants to create a 4-digit PIN code. This PIN is formed using digits from the set $\{0, 1, 2, \ldots, 9\}$.

(a) How many ways are there to create the PIN if digits can be repeated?

(b) How many ways are there to create the PIN if the digits must be distinct?

**Exercise 3.** A non-leap year has 365 days. Suppose January 1st is a Monday.

a) How many Sundays are there in this year?

b) How many Mondays are there in this year?

**Exercise 4.** A train consisting of 5 carriages stops at a station for 3 passengers to board. How many ways can these 3 passengers board the train if:

a) Anyone can board any carriage?

b) Each person boards a different carriage?

**Exercise 5.** Prove the following equations:

a) $C_n^0 = C_n^n = 1$ for all $n \in N$.

b) $kC_n^k = bC_{n-1}^{k-1}$ for all $n \in N$.

c) $C_n^k + C_n^{k-1} = C_{n+1}^k$ for all $n \in N$ and k = 1,2,...,n.

d) $C_n^k = \frac{n-k+1}{k} C_n^{k-1}$ for all $n \in N$ and k = 1,2,...,n.

**Exercise 6.** Choose 10 arbitrary natural numbers. Prove that there always exists a subset of these 10 numbers such that the sum of its elements is divisible by 10.

**Exercise 7.** Given 5 arbitrary points located inside a square with a side length of 2 units. Prove that there always exist at least two points among them whose distance is at most $\sqrt{2}$.

**Exercise 8.** Suppose $S$ is a finite set and $x \in S$ is an element of $S$. Prove that the number of subsets of $S$ containing $x$ is equal to the number of subsets of $S$ that do not contain $x$.

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